

Schedule of Meetings and Assignments or of Modules							
Module	Topic	Date	Subtopics	Expected Learning Outcomes (or You are expected to have understood/learned:)	In-class Activity	BEFORE CLASS	AFTER CLASS (ASSESSMENTS)
Mod 0	Introduction and Syllabus negotiation	9-Jan	Get to know each other. Set expectations between you (the students) and me (the instructor). Negotiation and communication skills. Self-assessment skills	Self-assessment skills. OR 6205 course structure. OR 6205 course agreements.	Students and Instructors presentations. Conversation of the class structure, the syllabus and expected learning outcomes.	Read Syllabus and this course map Take a look at Canvas setup and write down doubts	Pre-assessment quiz Recommended readings and videos shown in the Pre-assessment quiz
Mod 1	Introduction to Operations Research & Linear Programming	13-Jan	OR: why do we use it? Present an overview of OR framework. Decision Making, models, and who makes the decision. Modeling Strategies	Learners will be able to show understanding by: - Explaining how models are a product of context and decisions. Learners will be skilled at: - Documenting assumptions made during formulation - Critically evaluating whether a model adequately represents the intended situation - Critically identifying limitations and potential biases in formulations	Mini-lecture on OR framework Working on your decision problems. Working on Example problems	Chapters 1, 2	Upload Teams for the Project assignment Upload in-class exercises
	Generalized LP Model and solving it by hand	16-Jan	Components of a linear program: decision variables, objective function, constraints, and their standard forms. Geometric interpretation of feasible regions, extreme points, and optimal solutions. Steps of the graphical method for two-variable LPs.	Learners will be able to show understanding by: - Explaining geometric interpretation of feasible region and optimal solutions. - Identifying unbounded, bounded, infeasible and feasible LP problems in the graphical solution. Learners will be skilled at: - Identifying decision variables from problem descriptions - Writing objective functions that capture stated goals - Expressing constraints mathematically from verbal descriptions - Solving two-variable LPs graphically with clear reasoning	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Sections 3.1 - 3.3 & IOR Tutorial Mini-video on graphical representation of set of equations.	Upload in-class exercises
		20-Jan	Gauss-Jordan and solving system of equations, aka LP models	Learners will be able to show understanding by: - Explaining algebraic interpretation of feasible region and optimal solutions. - Identifying unbounded, bounded, infeasible and feasible LP problems in the algebraic solution. Learners will be skilled at: - Solving LP using Gauss-Jordan - Formulating LP models in non-standard form	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Chapters 3.4 - 3.5 & Appendix 4 Mini-lecture on algebraical Simplex	Upload what modeling tool your team chosen to use for this course Upload in-class exercises
Mod 2	Introduction to Simplex	23-Jan	Simplex method operations: pivot selection, tableau updates, optimality and feasibility conditions	Learners will be able to show understanding by: - Explaining how simplex moves and compare it with geometric solutions. Learners will be skilled at: Executing simplex iterations accurately by hand. Using Simplex to solve LP problems.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Chapter 4.1-4.4 and 4.9-4.10 Mini-lecture on simplex tableau	Upload in-class exercises
		27-Jan	Simplex Method and the use of software to solve bigger problems.		Game using Python at Google Colab - Computer required	Choose what to read and watch (1, 2 or 3) 1) Preferable - Python + Gurobi: Tutorial Pdf + Python Tutorial video 2) LINGO: Chapters 3.6, Supplemental Material Ch 3 + LINGO demo video 3) AMPL: Pdf Tutorial + Video Tutorial	Case study - Upload one of your preferable modeling tool Upload in-class exercises
Mod 3	Simplex, continued: Breaking Ties and Dealing with Non-Standard Forms	30-Jan	Breaking Ties and Degeneracy + Review	Learners will be skilled at: - Identifying non-standard situations and how to deal with it	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Chapter 4.5-4.6 Mini-lecture on simplex - Degeneracy	Upload in-class exercises
		3-Feb	Big-M and two-phase methods for handling non-standard forms	Learners will be skilled at: - Applying Big-M and two-phase methods when needed. - Identifying non-standard situations and how to deal with it	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Chapter 4.7-4.8 Mini-lecture on simplex - Big-M Mini-lecture on simplex - Two-Phase simplex	Upload in-class exercises
	Exam #1	6-Feb	Module 1 -3				
Mod 4	Theory of the Simplex Algorithm	10-Feb	Fundamental Insight	Learners will be able to show understanding by: - Showing how the simplex method moves along extreme points. Learners will be skilled at: - Using Matrix operations to get to the optimal solution.	Questionnaire Discussion groups	Read Chapter 5.1-5.3 & Appendix 2 Mini-lecture on simplex - Fundamental Insight	Project idea 1-page Upload in-class exercises

		13-Feb	Fundamental Insight and Computational complexity classifications and their implications.	Learners will be able to show understanding by: - Identify when one should use or could not use Graphical, Simplex by hand, Modeling language, Excel.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Chapter 5.4-5.5	Upload in-class exercises
Mod 5	Duality Theory	17-Feb	The relationship between primal and dual linear programs. Fundamental insight connecting primal and dual solutions	Learners will be able to show understanding by: - Identify the Dual and Primal of any LP problem - Showing relationship between the superoptimal and suboptimal relationship in primal and dual problems. Learners will be skilled at: - Transforming Primal to Dual	Questionnaire Discussion groups Example solved in class	Read Sections 6.1, 6.2, 6.3	Course Feedback Upload in-class exercises
		20-Feb	Complementary slackness conditions Apply revised simplex and dual simplex algorithms Course Feedback	Learners will be able to show understanding by: - Explaining complementary slackness conditions. Learners will be skilled at: - Using Dual Simplex	Conversation on Course Feedbacks and Course path correction	Read Sections 6.4, 6.5, 8.1	Upload in-class exercises
Mod 6	Sensitivity Analysis	24-Feb	Sensitivity analysis - shadow prices and reduced costs.	Learners will be able to show understanding by: - Explaining how real-world parameters are uncertain. - Explain shadow prices and reduced costs in context. Learners will be skilled at: Performing sensitivity analysis and interpreting results	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Sections 7.1, 7.2, 7.7	Upload in-class exercises
		27-Feb	Sensitivity analysis - changing in parameters and variables	Learners will be able to show understanding by: - Critically defending how sensitivity matters as much as optimality. - Showing how solutions change when inputs change is often more valuable than the optimal solution itself. Learners will be skilled at: - Performing sensitivity analysis and interpreting results - Critically assessing robustness of solutions to parameter changes	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Section 8.1	Project check-in Upload in-class exercises
	Spring Break	3-Mar					
		6-Mar					
	Exam #2	10-Mar	Module 4-6				
Mod 7	Classes of network optimization problems and their relationships (transportation, assignment, transshipment, shortest path, maximum flow, minimum cost flow, minimum spanning tree)	13-Mar	Transportation and Transshipment Problems structure and algorithms	Learners will be able to show understanding by: - Identifying the type of LP problem. - Identifying problems structure and which class it is. - Selecting appropriate algorithm to solve it. Learners will be skilled at: - Formulating this type of problem.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Sections 9.1, 9.2 Video on Transshipment problem	Upload in-class exercises
		17-Mar	Transportation and Transshipment Problems structure and algorithms	Learners will be skilled at: - Formulating this type of problem.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Video on Transportation Simplex	Upload in-class exercises
		20-Mar	Assignment Problem and its Algorithm	- Using step-by-step appropriate algorithm to solve it.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Sections 9.3, 9.4, 9.5 Video on Assignment Problem	Upload in-class exercises
Mod 8		24-Mar	Network Optimization	Learners will be able to show understanding by: - Recognizing classes of network problems - Explaining that efficiency of an algorithm is relative to problem. Learners will be skilled at: - Formulating this type of problem.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Sections 10.1, 10.2, 10.3 Video on Network Optimization	Upload in-class exercises
		27-Mar		Learners will be able to show understanding by: - Identifying main structure of Network Problems. Learners will be skilled at: - Using step-by-step of Network Simplex to solve it.	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Read Sections 10.4, 10.5, 10.6	Upload in-class exercises
		31-Mar	Project Networks and Project Review	Establish Expectations for Course Project and have time to work on presentation for next class	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Chapter 10.8	Upload in-class exercises
	Good Friday	3-Apr					
	Project Presentation and Peer Feedback	7-Apr	Project: Communicate recommendations to non-technical audiences. Defend modeling choices.	Learners will be able to show understanding by: - Giving and receiving feedback - Communicating recommendations to decision-makers who may not understand mathematics, translating model outputs into clear business or operational guidance with appropriate caveats.	Presentation	Text How to give and receive effective feedback	Peer evaluation (online) Upload feedback to peers

Mod 9	Introduction to Dynamic Programming	10-Apr	Set up and solve dynamic programs using recursive reasoning.	Learners will be able to show understanding by: - Explaining the principle of optimality and structure of dynamic programming problems Learners will be skilled at: - Setting up and solving small dynamic programs	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises		Project draft Report due Upload in-class exercises
	Introduction to Dynamic Programming- Additional Applications	14-Apr		Learners will be able to show understanding by: - Creating the framework to solve a general DP problem Learners will be skilled at: - Setting up and solving small dynamic programs	Example solved in class Exercises in class (individual, peer) Class "debugging" of exercises	Video on Recursion	Upload in-class exercises
	Continuous State Dynamic Programming and Review	17-Apr	Project: document assumptions. Offer recommendation of action based on model results. Evaluate implications of modeling decisions.	Learners will be able to show understanding by: - Effectively using feedback to improve project decisions and report. Learners will be skilled at: - Setting up and solving small continuous state dynamic programs	Example solved in class Exercises in class peer Review for Exam	Video Continuous State Dynamic Programming	Project Report Upload in-class exercises
	Final Exam ? TBD later the specific day	21-Apr	Module 7-9				
		24-Apr					

Exam and Projects deadlines
NO CLASS